

ANOTHER DIMENSION

Disrupter: 4D printing, smart materials, 3D printing with mixed materials

Disrupted: Spare part dealers, simple manufacturing, China

What Year: 2025 to 2030

OK. So what the heck is 4D printing anyway. Some new fancy term for faster 3D printing? Not really. When we use the term, we mean 3D printed objects that can assemble themselves into more complex structures.

The materials used here are called smart materials, which isn't the traditional use of the word "smart" – the materials themselves don't think... but then again, neither does your phone or TV, and they're called smart too ;-)

A smart material is one that changes shape automatically to form a new shape. Shape memory alloys are a type of smart material, and they are able to remember their resting shape. When we say remember, we mean that they are able to revert to a particular shape given the same conditions of pressure, temperature, etc. It's not as simplistic as a football, which changes shape all the time when it is kicked, and then reverts to a near-spherical shape

because of the air pressure and the shape of the stitched leather. Here we mean a material which will take one shape to start with and then transform into a more complex shape. So, for, example, in MIT labs, they're 3D printing strings, which when put into a liquid, will transform and become anything they want – "MIT" written in cursive, the shape of a protein, etc. If you wanted to 3D print that shape, it would be complex, you'd need a much more expensive 3D printer, and it would be a slow printing process. With smart materials you can quickly print a flattish shape – say, a cardboard cutout-like cross – that folds up to become a cube. Printing a hollow cube would be a pain with a 3D printer, but a cross is easy. Then you leave it in a solution, and it folds itself to become a cube!

All of this is made possible by 3D printing, and some would say it's just all advancements in 3D printing. However, with more and more materials and

smart materials finding their way into 3D printing, that day is not far off when you will be able to print your own spare parts at home. Want a carbon fibre tennis racket? Print it! Want new shoes, but with your own personal design on them? Print those out too...

Marry the previous disrupter with this one, and you can see how it spells doom for all those little repair shops everywhere. Spare parts business? What would be the point of that? It might also be of global significance. You know all those Chinese things you buy – toys, gadgets, everything... – why buy them when you can print them yourself? It would just be cheaper in the long run...

If 3D printing catches on as it should, we're predicting the end of China's dominance over everyday household products that are shipped across the globe. They may still make all the iPhones, but you will print your own plastics for the home!

ANOTHER BRICK IN THE WALL

Disrupter: Internet

Disrupted: Education

What Year: 2020 onwards

We're predicting the end of education as we know it. Sure, maybe you will still go to school to get a formal education, but you may not have to learn as a "classroom", where everyone learns the same things. You may actually choose or be given the task of learning something very different from other kids your age. This will be a good thing. Also, expect to see more and more adults going back to school.

With fast and consistent access to the internet, most of us are already getting our knowledge solely online. A few of the videos and courseware from our Digit DVD and the Fast Track and dmystify might help a lot of you learn new things – and this is good because it pays to have someone choose what you should read about once in a while.

However, we're willing to bet that most of your reading and video watching is done online. You're usually looking for stuff based on what you need at the moment.

So what do you do if you want to take up a summer job at McDonald's? You get some wise-ass who has the IQ of a bug lecturing you about using the fryer. Wouldn't it be so much cooler for McDonald's to send you off to a room with a computer and a fryer where you attend an online course on how to fry, well, fries. You then demonstrate to some senior your fryer skills, and if satisfactory, get on with working and earning some pocket money. It's a win for McDonald's, because they get to ensure uniformity of the fries across locations, and for you because you don't have to be tortured by bug-boy.

It's not just for simple jobs like that... you could be an investment banker, and get a quick crash course in a new software product. It could be a Yoga course to help you relax, and control your blood pressure during intense stock trading sessions, leading to more cool-headed deals.

Just bought a drone from China? Take a quick crash course in how to assemble and program it. Literally anything you wanted to learn, could be taught to you.

Technology is set to change the education game in ways that they cannot imagine. Many forward-looking educational institutes are already investing in courseware and specific courses like the ones we illustrated above. They will probably be left standing when the dust settles.